Urban Microclimate Modelling

FAST WIND SURROGATES FOR CITY BUILDING LAYOUTS USING ML, DL AND A HYBRID

Can a **fast learned model** predict wind around building layouts accurately enough to stand in for slow CFD simulations, and **where does it fail**?

TEAM

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Project 6

CSAI Project
Newton School of Technology
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What we built and why

THE PROJECT

A surrogate model is a **fast stand-in** for CFD.

Phase 1 (ML) predicts one number per layout, the canopy mean wind speed.

Phase 2 (DL) predicts the whole wind map at pedestrian height.

Then we test a **hybrid** of both.

THE DATA

UrbanTALES dataset of large-eddy simulations of idealized building arrays.

224 simulations: 112 layouts × 2 wind directions across 16 layout families.

ML used a small **224-row** table.

DL used 224 pedestrian NetCDF velocity files plus 224 topography files, **~15.2 million** air cells in total.

THE MOTIVATION

CFD takes **hours** or **days** per layout.

Street wind affects **comfort, heat and air quality.**

The brief gives a natural **ML-to-DL** story.

Wind maps and error maps are **easy to check visually.**

A ready dataset let us **focus on modelling.**

Phase 1 recap

Machine Learning pipeline, results and the limit

Notebooks ML_01 to ML_07
data dictionary, EDA, leakage audit and features

↓

Family-held-out split
whole layout families held out

↓

Models trained and compared
Ridge, Random Forest, XGBoost

↓

Evaluation
fair test with 5-fold GroupKFold

BEST MODEL

XGBoost

0.232 m/s | **0.90 R-squared**

Other models

Random Forest	0.256
Ridge	0.400
Mean baseline	0.798

Density features carry about **73%** of XGBoost importance

THE LIMIT

A few summary numbers cannot show **wakes or street-canyon effects.** So we move to maps.

Phase 2

Why deep learning and key research

Deep learning sees the layout as an **image** and outputs a **full wind map.**

U-Net (Ronneberger 2015) gives an encoder-decoder with skip connections — **our backbone.**

Thuerey et al. 2020: a U-Net predicts flow fields and **generalises to unseen shapes.**

Roberts et al. 2017 shows random splits overstate accuracy; we hold out whole **layout families.**

Chen and Guestrin 2016 (**XGBoost**) is our Phase 1 model and the **hybrid channel.**

Nazarian et al. 2025 provides **UrbanTALES**, our dataset.

DL_01 and DL_02

DL_01

Suhaani Garg

reads velocity (uped, vped) and building maps for all **224 cases**

valid-air mask created

no resampling, **0.5 m** grid kept

normalisation fitted on training cases only (7.**68 million** air cells)

DL_02

Aarya Srivastava

compact U-Net with **486,706 parameters**

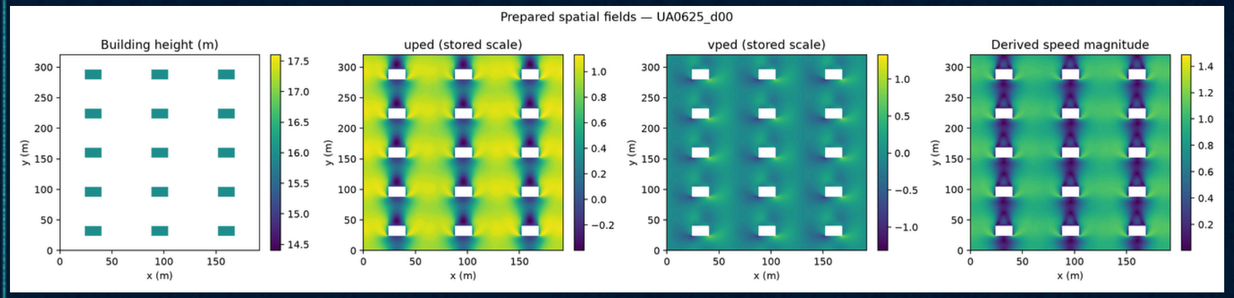
5 input channels: building height, wind direction sin and cos, friction velocity, mask

2 outputs: u, v

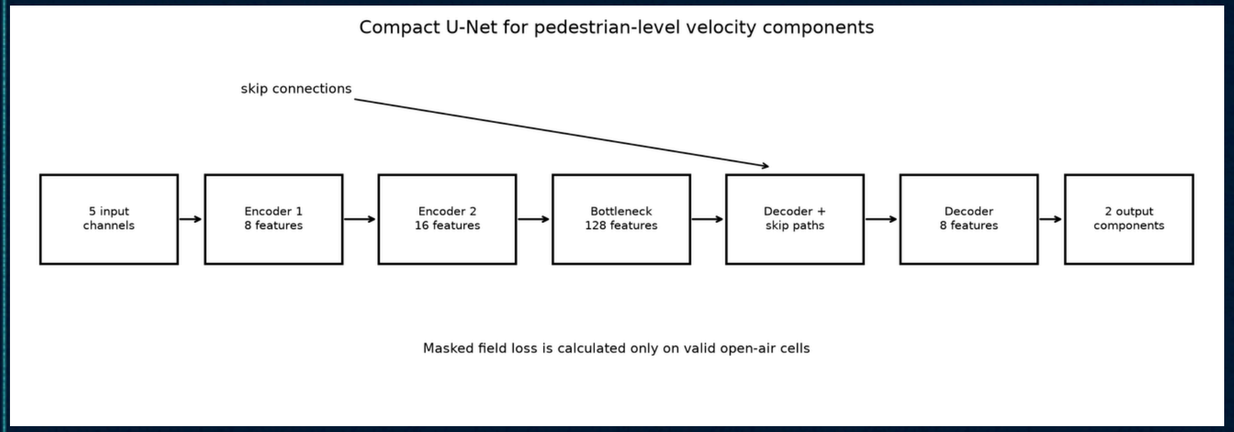
masked MSE loss ignores buildings

small CNN baseline with **32,570 parameters**

DL_01: PREPARED DATA, CASE UA0625_D00



DL_02: COMPACT U-NET ARCHITECTURE



DL_03 and DL_04

DL_03 (PALAK)

Families split
10 train (140 cases)
3 validation (42)
3 test (42)

Training setup
AdamW, lr 0.001, wd 0.0001
gradient clipping 1.0, dropout 0.1
early stopping, seed 42

DL_04 (JASHVITHA LAKSHMI OMKARAM)

Tested on unseen families
A0625, A1111, A4444
(5.16 million air cells)

U-Net error by family
0.300 (A0625 sparse)
0.161 (A1111)
0.081 (A4444, dense)

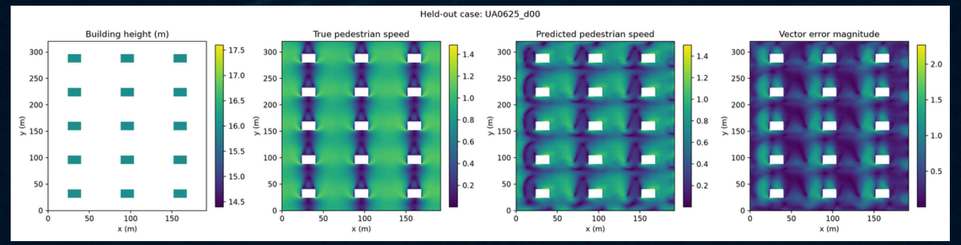
TEST RESULTS (FIELD MAE)

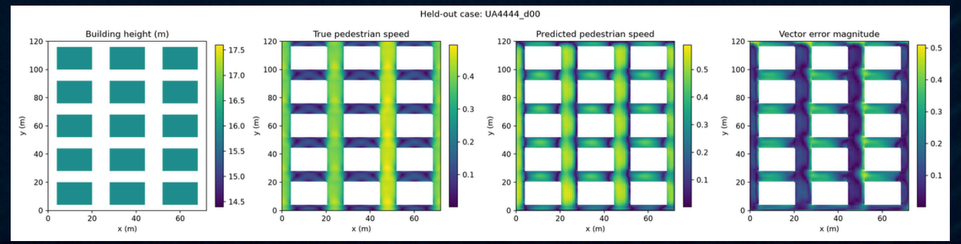
U-Net	**0.2438**
compact CNN	**0.2238**
geometry-only ablation (10 epochs)	**0.2815**
no friction velocity ablation (10 epochs)	**0.2623**

HONEST NOTE: the simple CNN beat the U-Net on test; ablations used 10 epochs vs 25.

DL_03: U-NET LEARNING CURVE

DL_04: HELD-OUT ERROR MAPS





Hybrid model experiment

DL_05 | Samarth Sangtani

Phase 1 XGBoost canopy-speed prediction fed to U-Net as a **6th constant channel.**

Leak-free training and testing.

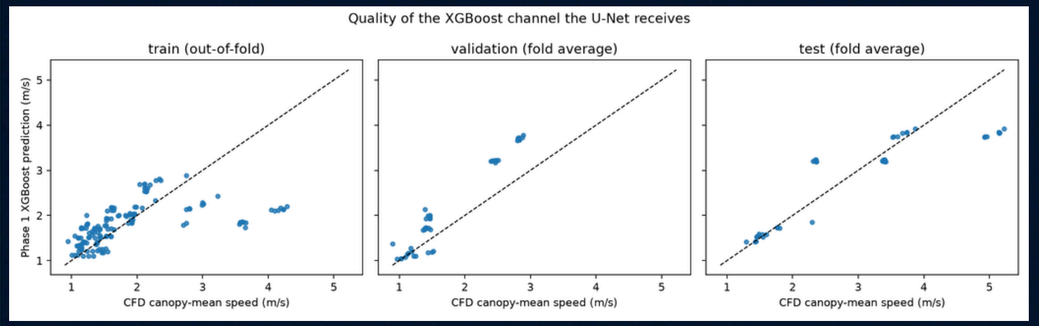
Shuffled-channel control added.

MODEL VARIANT	TEST MAE
Base U-Net	0.2438
Hybrid (XGBoost + U-Net)	**0.2527**
Shuffled-channel control	**0.2386**

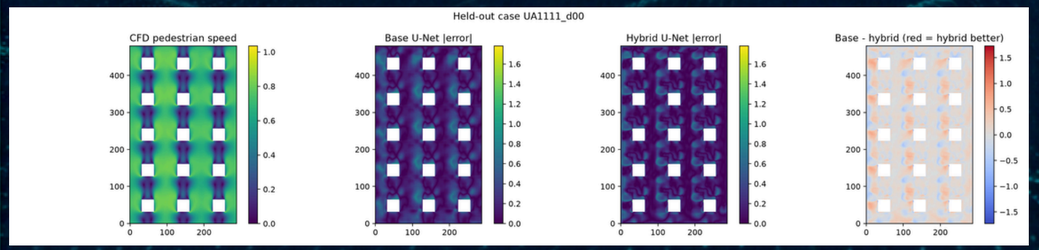
Hybrid better on only **16 of 42** test cases.

No measurable gain.

XGBoost channel the U-Net receives



Base vs hybrid error, case UA1111_d00



Comparison, conclusions and final solution

PHASE 1 — AVERAGE CANOPY WIND SPEED

XGBoost	0.232
Random Forest	0.256
Ridge	0.400
baseline	0.798

PHASE 2 — FULL WIND MAP

compact CNN	0.2238
U-Net	0.2438
hybrid	0.2527

Phase 1 and Phase 2 errors are not directly comparable

WHAT WE CONCLUDE

density drives average canopy wind.

U-Nets give wind maps for unseen layouts.

the **hybrid channel** did not help.

FINAL RECOMMENDED SOLUTION

use
XGBoost
for a quick average canopy wind speed

plus
compact CNN
or **U-Net**
for the wind map

leave the **hybrid** out

NEXT | repeat with **3 seeds** and give ablations the **same training budget**.

Team contribution

Clear ownership across data, models and validation.

NOTEBOOK	TEAM MEMBER	RESPONSIBILITY
ML_01	Suhaani Garg	dataset understanding and data dictionary
ML_02	Aarya Srivastava	EDA and target definition
ML_03	Jashvitha Lakshmi Omkaram	leakage audit and feature engineering
ML_04	Jashvitha Lakshmi Omkaram	grouped split and preprocessing
ML_05	Palak	Ridge regression
ML_06	Samarth Sangtani	Random Forest and XGBoost
ML_07	Suhaani Garg	evaluation, comparison and diagnostics
DL_01	Suhaani Garg	dataset audit and data preparation
DL_02	Aarya Srivastava	literature review and architecture
DL_03	Palak	training and regularization
DL_04	Jashvitha Lakshmi Omkaram	validation, error maps and ablations
DL_05	Samarth Sangtani	hybrid ML + DL integration

Thank you

Code and results:

github.com/aarya-sr/T006-urban-microclimate-modelling